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> # Атака с малой экспонентой
> with(numtheory) :
> with(RandomTools) :
> e := 3;
                                     e := 3
(1)
> p1 := nextprime(Generate(integer(range=2..10^5)));
                                     p1 := 40697
(2)
> q1 := nextprime(Generate(integer(range=2..10^5)));
                                     q1 := 73597
(3)
> n1 := p1·q1;
                                     n1 := 2995177109
(4)
> p2 := nextprime(Generate(integer(range=2..10^5)));
                                     p2 := 13619
(5)
> q2 := nextprime(Generate(integer(range=2..10^5)));
                                     q2 := 65551
(6)
> n2 := p2·q2;
                                     n2 := 892739069
(7)
> p3 := nextprime(Generate(integer(range=2..10^5)));
                                     p3 := 19391
(8)
> q3 := nextprime(Generate(integer(range=2..10^5)));
                                     q3 := 2351
(9)
> n3 := p3·q3;
                                     n3 := 45588241
(10)
> igcd(n1, n2);
> igcd(n1, n3);
> igcd(n2, n3);
                                     1
                                     1
                                     1
(11)
> igcd(n1, e);
> igcd(n2, e);
> igcd(n3, e);
                                     1
                                     1
                                     1
(12)
> m := ((Generate(integer(range=2..10^10)) mod n1) mod n2) mod n3;
                                     m := 34094331
(13)
> y1 := m&^e mod n1;
> y2 := m&^e mod n2;
> y3 := m&^e mod n3;
                                     y1 := 1877344059
                                     y2 := 745149195
                                     y3 := 42574485
(14)
> c := chrem([y1, y2, y3], [n1, n2, n3]);
> M := root(c, e);

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$m - M;$

$c := 39632048377820302466691$

$M := 34094331$

0

(15)